Ghosting in MR Images: Theory

“Ghosting or phase smearing is one of the most common image quality complaints in Magnetic Resonance Imaging. Typically, it appears as multiple replicas of the scanned object spread across the phase-encoded axis, or the scanned object will appear smeared across the phase-encoded axis.

Ghosting may also be caused by motion of tissue or motion of the patient during the scan. This type of ghosting is mostly understood by technologists and radiologists, but in certain cases may be mistaken for a system failure. Although this is an applications type problem, it is covered briefly in the flowchart to help the Field Engineer avoid spending time troubleshooting a problem that he or she can't repair. The "Other Artifacts" section of the main menu also shows some examples of ghosting type artifacts that can't be repaired by adjustments or hardware replacement.
The root cause of hardware induced ghosting in images is shifts in Echo Center, Constant Phase, or Magnitude instabilities of the echoes that are acquired to produce the image. Regardless of the source or type of instability, the resultant ghosting or smearing will always appear along the phase-encoded axis of the image. Possible sources of instability in the system are many. In fact, most subsystems have the potential to produce instabilities that will cause ghosting in images. Although there are many different types and sources of instabilities, the ghosting they produce in images looks similar.

There are three basic types of instability: Magnetic, RF Transmit, and RF Receive. **Magnetic** instabilities are those that affect the magnetic field (Bo) and/or the magnetic gradients used for spatial encoding. Instabilities of this type usually affect phase and frequency stability. Possible sources of this type of instability are, the gradient subsystem, shim supplies, shield coolers, thermal acoustic oscillation and vibration. **RF transmit** are those instabilities that affect the B1 or RF field. Instabilities of this type usually affect magnitude and phase stability. Some possible sources for this type of instability are, the RF amplifier, the envelope feedback assembly, the IPG and any of the components of the RF transmit chain. **RF receive** instabilities are those that affect the received echo or signal. Instabilities of this type affect magnitude stability. Some possible sources of this type of instability are, the TPS receiver, the preamplifier and any of the components of the receive chain.

Since the appearance of ghosting in images looks similar regardless of the source, it is inappropriate to try to troubleshoot the problem from clinical images. For that reason, SST test data will be used to troubleshoot throughout the ghosting flowchart.”